UNIT 6 / DAY 15 / HONORS

# Radar range and velocity

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Separate a radar's range measurement from its line-of-sight velocity measurement.

**SUCCESS CRITERIA - I can...**

1. Use round-trip time for range.
2. Explain the information in a Doppler shift.
3. State what radial velocity leaves unknown.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

An EM wave in vacuum has frequency **$5\times10^{9}$ Hz**.
1. Use c=**$3.00\times10^{8}$ m/s**.
2. **Find wavelength** and **name the spectral region**.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** range = c Δt/**2**; c = **$3.0\times10^{8}$ m/s**.
Radar time is a round trip. Doppler gives line-of-sight velocity information.

**Worked reasoning:** A radar echo returns after **2 microseconds**. With c=**$3\times10^{8}$ m/s**, range=c Δt/**2**=**300 m**. The measured time includes the outward and return paths.

**Your turn:** Draw the round trip and **label the conversion 2 microseconds=$2\times10^{-6}$ s**.




**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

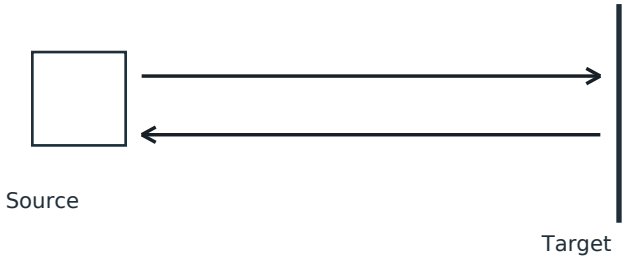
1. **Draw outgoing** and returning radar paths.
2. **Which measurement** helps range: echo delay or frequency shift?
3. Which helps radial velocity?




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A stationary radar receives an echo after **0.040 s**. Use c = **3 × $10^{8}$ m/s** to **find range**. Separately, its wavelength is **0.10 m** and Doppler-shift magnitude is **400 Hz**. Use $|v_r| = |\Delta f|\lambda/2$ to **find radial speed**. **Label which measurement supports each** result.




**Marzano check - circle:** ① ② ③

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## 05 Explain your reasoning

An asteroid has zero measured radial velocity at one instant. Does that prove it is stationary? **Sketch another possible motion**.

Add a velocity perpendicular to the line of sight.




**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A wave travels **12 m** in **0.4 s** at frequency **60 Hz**. **Find speed** and wavelength.
**Label a wave sketch** to **support the calculation or comparison**.




Time or position: choose and label

**Marzano check - circle:** 1 2 3

## 07 Exit check

Two targets at the same range have different Doppler shifts. **Explain why they can have** same time-delay but different motions.




**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

Radar range changes from **1500** to **1440 m** over **0.5 s**.
1. **Find average radial velocity taking** outward positive.
2. **Explain why total speed may** be larger.




**Marzano check - circle:** 1 2 3